

LETTER TO THE EDITOR

WILEY

Spontaneous regression of tumor in a girl with optic pathway glioma with diencephalic syndrome

To the editor:

A 6-month-old girl presented with poor appetite but no vomiting or other gastrointestinal symptom. There was no significant increase in her body weight for 2 months. Her birth weight was 3.4 kg. She also had horizontal nystagmus 2 days before her initial visit. Physical examination showed that the patient had bilateral horizontal nystagmus and left esotropia. Poor visual acuity was found on the left side, but good acuity on the right. No café-au-lait macule was found on her body. Visual activity examination and flash visual evoked potential (FVEP) by the ophthalmology department failed. Magnetic resonance imaging (MRI) demonstrated a 42 mm × 37 mm × 36 mm enhanced and lobulated lesion with a large cystic component at the bottom located in the sellar region with a clear boundary with surrounding tissues (Figure 1A, B). The optic chiasm and optic nerve nearby could not be recognized. Optic pathway glioma (OPG) and diencephalic syndrome (DS) were diagnosed according to the history and the characteristics of the MRI. No family history of tumor including neurofibromatosis type 1 (NF-1) was found. Considering the low age and low body weight of the patient, her parents refused surgical treatment. Close follow up (3 months) was suggested.

One month later, horizontal nystagmus of the patient disappeared. However, her body weight declined continuously and came to the lowest point of 4.5 kg at the age of 1-year-old. The patient's appetite and food intake increased gradually after the age of 1 year. The MRI of the patient at 1 year and 11 months of age showed the size of the tumor reduced to 38 mm × 35 mm × 23 mm (Figure 1C, D). A newly generated cystic component that seemed like an arachnoid cyst formed at the left temporal pole.

The girl grew to 10 kg at the age of 2 years and 2 months. She seemed to recover from DS. Her parent did not find a remarkable decrease in her vision and horizontal nystagmus

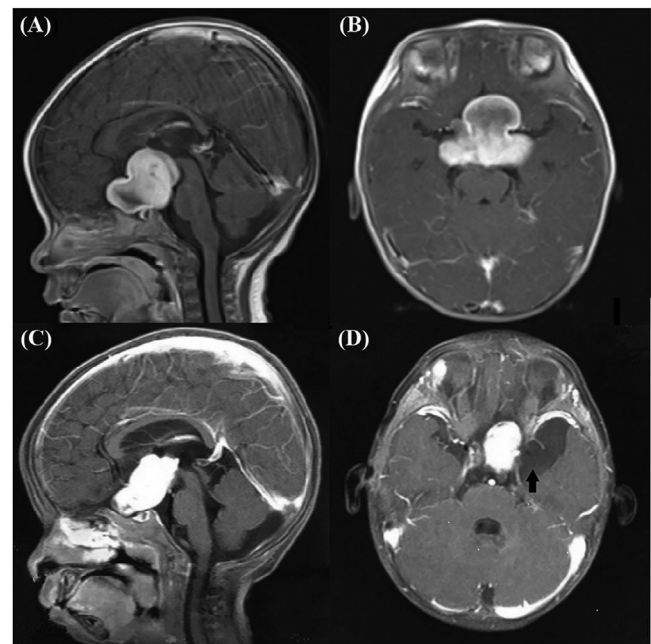


FIGURE 1 Magnetic resonance imaging of a girl with optic pathway glioma and diencephalic syndrome. (A) Sagittal and (B) axial enhanced MRI images of the patient at her age of 6 months and 20 days. (C) Sagittal and (D) axial MRI images of the patient at her age of 1 year and 11 months old. Tumor size shrunk obviously from 42 mm × 37 mm × 36 mm to 38 mm × 35 mm × 23 mm. A newly generated cystic component that seemed like an arachnoid cyst (18 mm × 35 mm × 26 mm) formed at the left temporal pole.

did not appear again. Visual activity examination and flash visual evoked potential showed she had a moderate retardation of her left eye and a normal signal of the right eye.

Optic pathway gliomas are special gliomas originating in the visual pathway, usually found in children under 10 years of age. The biological behaviors of OPGs are unstable. Some can remain stable for a long time but some grow

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or spread rapidly, and several even shrink or disappear naturally.^{1–3} Optic pathway gliomas also lead to various endocrine changes, including DS.⁴ Diencephalic syndrome can lead to difficulty in gain weight in pediatric patients, usually secondary to neoplastic lesions nearby hypothalamus, was closely related to the OPGs.⁵ Some studies have shown that OPG patients with DS have a poor prognosis. Gropman et al.⁶ found that the average survival time of untreated OPGs patients with DS was usually less than 1 year.⁶ Rakotonjanahary et al.⁷ reported that girls with OPG and DS had a poor prognosis. In this case, without any treatment, the tumor size reduced and DS improved.

Natural regression of tumors including central nerve system tumors has been reported. Brzowski et al.⁸ first reported a 2-year-old boy with NF-1 who developed a natural regression of optic glioma. Natural regression was mostly observed in patients with NF-1.³ However, Parsa et al.⁹ reported that 13 children with optic glioma had natural regression, nine of whom did not have NF-1. The mechanism of natural regression of OPG has not been clearly studied. Some researchers thought that it was mostly due to the shrinking of cystic components. However, some studies had shown tissue regression of astrocytoma located in several positions.¹⁰ Some believed reduced proliferation could be secondary to enhanced immunocompetence and immune recognition of tumors.⁹

Treatment for OPG patients with DS is still controversial. Parsa et al.^{11,12} thought that OPGs should be categorized as hamartomas rather than neoplasms; therefore they should not be treated, even if there is an evidence of clinical or radiographic progression. However, some believed this kind of tumor still belongs to the neoplasm.¹³ Thiagalingam et al.¹⁴ showed that the reduction of tumor size was not always positively correlated with the patients' visual function; the vision of some OPG patients with natural regression could remain stable or even deteriorate. During the follow up of this case, we found the original nystagmus of the patient disappeared and the emaciation caused by DS gradually improved, but the left eye vision did not improve. We thought that body weight may reflect the progression of OPG patients with DS better than visual activity. As far as we are aware, this is the first case in which spontaneous regression occurred in an OPG patient accompanied by DS. Even though patients with DS always showed a worse outcome, for children with recent weight gain, close follow up may be a choice.

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CONSENT FOR PUBLICATION

Written informed consent was obtained from the parents.

CONFLICT OF INTEREST

The authors declare no competing interest.

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